

External Interface Validation Summary

NCAOS Boundary Signals × Urielle Audit Layer

PoC Evaluation — Interface Sufficiency Test

Reference Implementation

A lightweight PoC implementation is available for reference:

<https://codesandbox.io/p/sandbox/n4cfst>

Objective

Validate whether boundary-emitted governance signals can be consistently interpreted, mapped into control domains, and translated into audit artifacts without requiring internal system access.

Input (Illustrative Event Set)

Event Type	Action Taken	Layer
policy_violation	blocked	boundary
unauthorized_access	denied	boundary
state_anomaly	restricted	boundary
integrity_breach	shutdown	boundary

Event → Audit Mapping

Event	Signal	Control	Outcome	Evidence
policy_violation	Boundary breach	Policy	Blocked	Intervention log
unauthorized_access	Unauthorized access	Access Control	Denied	Access log
state_anomaly	Deviation	Stability	Restricted	Deviation record
integrity_breach	Integrity failure	Integrity	Shutdown	Alert log

Key Observations

- Signals are semantically clear and sufficient for interpretation.
- Events map cleanly into control domains.
- Audit artifacts can be generated without internal system access.
- Independence between control and audit layers is preserved.

Conclusion

The boundary-emitted event schema is sufficient for consistent interpretation, control mapping, and audit artifact generation while preserving architectural independence.

Key Outcome

NCAOS signals can serve as a stable interface for external audit systems without requiring internal system access.